

Natural Language Processing

Lab 3: Sentences - parsing style and structure

2025/26

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1. For this week's lab session we will work with a set of US presidential speeches: inaugural addresses
 - (a) Download the corpus from the Moodle and extract the zip file
 - (b) Write python code to read in the text files from this folder. Here is one example of how to read files from a directory:

```
import os
import glob

# Specify the directory containing the .txt files
directory_path = 'path/to/your/directory'

# Use glob to find all .txt files in the specified directory
txt_files = glob.glob(os.path.join(directory_path, '*.txt'))

# Read the contents of each .txt file
for txt_file in txt_files:
    with open(txt_file, 'r') as file:
        content = file.read()
        print(f"Contents of {txt_file}: \n{content} \n")
```

- (c) What parameters does the open function take in python? Which might be relevant here?
2. In the lecture we saw that the type-token ratio can give a simple estimate of text difficulty. Calculate the type-token ratio of each text.
 3. Follow the <https://spacy.io/usage/linguistic-features> for extracting linguistic features using spaCy
 4. Parse the texts using spaCy and produce the following outputs:
 - (a) The named entities in each text
 - (b) The adjectives in each text
 - (c) The most common adjectives overall.

(d) The most common adjective-noun pairs

5. Using only `nltk`, construct a word-word co-occurrence matrix that counts how often each word occurs in the same sentence as each other word. Can you use this to construct a pointwise mutual information score for each word pair?